

CHITA KADALA, Russian Federation

WMO: 307580

Lat: 52.026N

Lon: 113.306E

Elev: 2272

StdP: 13.53

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.9	-31.2	-41.5	0.5	-34.9	-36.7	0.7	-30.6	21.8	6.1	19.0	6.7	0.4	60	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	88.0	65.2	84.4	63.9	80.9	62.3	69.1	81.9	66.8	78.2	64.8	75.6	7.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	99.8	74.0	62.8	93.1	71.6	60.9	87.0	69.8	34.8	82.9	32.8	78.4	31.2	75.1	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.0	18.8	16.2	DB	-40.0	93.9	5.1	4.1	-43.6	96.8	-46.6	99.2	-49.5	101.5	-53.2	104.5	(4)
(5)			WB	-40.1	72.6	5.0	3.9	-43.7	75.4	-46.7	77.7	-49.5	80.0	-53.1	82.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.0	-10.8	0.0	17.5	36.1	49.8	62.8	67.3	62.4	49.5	32.4	10.1	-6.8	(6)	
	DBStd	28.60	10.84	11.10	11.23	8.39	7.57	6.57	5.05	5.71	7.16	8.63	12.26	11.91	(7)	
	HDD50	8416	1883	1399	1007	422	98	3	0	2	96	547	1197	1762	(8)	
	HDD65	12596	2348	1819	1472	868	473	114	32	118	467	1011	1647	2227	(9)	
	CDD50	1488	0	0	0	5	92	388	535	387	80	1	0	0	(10)	
	CDD65	192	0	0	0	0	2	49	102	38	1	0	0	0	(11)	
	CDH74	2962	0	0	0	4	143	941	1285	535	54	0	0	0	(12)	
	CDH80	970	0	0	0	0	36	321	461	144	8	0	0	0	(13)	
(14) Wind	WSAvg	5.7	3.5	4.4	6.1	8.1	8.3	6.2	5.4	5.2	6.0	6.2	5.1	3.6	(14)	
(15) Precipitation	PrecAvg	13.5	0.1	0.1	0.1	0.4	1.0	2.3	3.7	3.3	1.6	0.4	0.2	0.2	(15)	
	PrecMax	24.5	0.3	0.4	0.6	2.2	3.0	7.3	13.0	9.3	4.8	1.9	0.6	0.7	(16)	
	PrecMin	7.8	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.7	0.1	0.0	0.0	0.0	(17)	
	PrecStd	3.6	0.1	0.1	0.1	0.4	0.6	1.3	2.2	1.5	1.0	0.4	0.2	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	34.9	53.5	71.3	85.0	93.1	94.9	88.1	80.3	63.6	44.2	28.4	(19)	
		MCWB	20.4	28.1	40.0	48.9	60.7	66.7	70.5	65.4	59.3	47.5	35.6	23.9	(20)	
	2%	DB	16.0	28.4	46.5	63.0	77.3	87.6	89.3	84.1	73.7	56.9	37.7	22.6	(21)	
		MCWB	13.3	23.3	35.2	45.9	55.4	64.5	67.1	64.6	56.6	43.6	31.2	19.4	(22)	
	5%	DB	10.7	23.2	41.2	57.6	72.7	83.8	84.8	80.2	68.6	51.2	32.9	17.6	(23)	
		MCWB	8.7	19.4	31.8	43.0	52.5	63.0	64.8	63.3	54.0	40.1	27.6	15.2	(24)	
	10%	DB	6.4	18.7	35.9	52.9	67.8	79.4	81.1	76.0	64.2	46.4	28.4	12.1	(25)	
		MCWB	4.7	15.5	28.5	40.1	50.1	60.5	63.5	61.7	52.2	37.5	24.4	10.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	27.9	40.8	51.1	62.6	70.8	74.4	70.8	62.5	48.4	36.4	24.0	(27)	
		MCDB	23.7	34.1	52.6	66.7	80.5	87.7	90.8	83.3	75.5	62.7	43.5	28.0	(28)	
	2%	WB	13.3	23.2	35.6	46.8	57.6	67.8	70.2	67.5	59.1	44.1	31.4	19.2	(29)	
		MCDB	15.9	28.0	45.8	61.3	74.1	82.5	82.6	77.7	68.2	54.4	37.4	22.3	(30)	
	5%	WB	8.8	19.5	31.9	43.5	54.0	64.6	67.6	65.1	56.3	40.9	27.9	15.0	(31)	
		MCDB	10.8	23.3	40.6	57.1	68.9	78.0	79.1	74.6	66.1	49.6	32.5	17.5	(32)	
	10%	WB	4.5	15.3	28.7	40.4	50.9	62.2	65.6	63.1	53.4	37.9	24.4	10.1	(33)	
		MCDB	6.0	18.3	35.8	52.1	65.5	75.4	76.3	72.7	62.5	45.6	28.3	11.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	22.0	28.0	27.6	25.4	27.1	27.0	24.3	23.4	25.2	22.6	20.8	20.2	(35)	
		MCDBR	22.9	30.3	31.2	34.8	37.9	35.7	32.6	31.7	34.1	30.2	23.1	21.7	(36)	
	5% WB	MCWBR	20.8	25.9	21.9	21.6	20.2	16.9	14.4	15.2	18.5	18.5	18.0	19.1	(37)	
		MCWBR	23.1	30.1	30.9	33.0	34.5	30.8	26.3	26.4	30.3	27.9	22.7	21.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.252	0.280	0.320	0.407	0.508	0.478	0.426	0.394	0.347	0.304	0.276	0.256	(40)	
		taud	2.208	2.173	2.155	2.065	1.834	2.031	2.252	2.323	2.394	2.425	2.285	2.167	(41)	
	Ebn at Noon		239	264	274	261	238	248	261	264	265	256	228	213	(42)	
		Edh at Noon	21	30	37	47	62	51	41	36	30	23	20	19	(43)	
(44) All-Sky Solar Radiation	RadAvg		454	817	1327	1654	1832	1929	1780	1520	1195	790	460	343	(44)	
	RadStd		13	18	44	118	115	135	149	100	89	49	24	17	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.76	-1.91	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page